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ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Course: B. Tech (ME)

Name of the Paper: Mechatronics

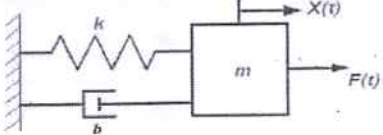
Time: 3 Hours

Semester: **VII**Paper Code: **TME 708**

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	What do you understand by a mechatronic system? Give examples of mechatronic systems.	CO1
(b)	What is a control system? Describe the types of control systems.	
(c)	What are the different signal conditioning processes?	
Q2	(20 marks)	
(a)	Describe the pneumatic power supply system with diagram.	CO2 & CO3
(b)	Explain the working of a solenoid with diagram.	
(c)	What are the different types of brushed DC motor? Explain with diagrams.	
Q3	(20 marks)	
(a)	Describe the working of a LVDT sensor with diagram.	CO4
(b)	What is the electrical resistance strain gauge? Explain with diagram.	
(c)	What are the different pressure measurement devices?	
Q4	(20 marks)	
(a)	Describe the structure of a microcontroller with diagram.	CO5
(b)	Describe the programmable logic controller.	
(c)	Describe the inverting and the non-inverting types of the operational amplifiers with their governing equations.	
Q5	(20 marks)	
(a)	What do you understand by the transfer function? How to describe the stability of a system using the characteristic equation?	CO6
(b)	Derive the transfer function of the following mass-spring-dashpot system.	
		
(c)	Derive the differential equations of the following electrical R-L-C circuit. Consider input is e_i and output is e_o .	
	